



An-Najah National University
Computer Engineering Department
Distributed Operating Systems

Bazar.com: A Multi-tier Online Book Store

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The project Hierarchy:

BAZAR.COM/

```
├── node_modules/
├── src/
│   ├── catalog/
│   │   ├── catalog.js
│   │   ├── catalog.json
│   │   ├── Dockerfile
│   │   └── package.json
│   ├── front-end/
│   │   ├── Dockerfile
│   │   ├── frontend.js
│   │   └── package.json
│   └── order/
│       ├── Dockerfile
│       ├── order.js
│       ├── orders.json
│       └── package.json
├── docker-compose.yml
├── package-lock.json
├── package.json
└── README.md
```

Bazar.com is a microservices-based system with three components:

- Catalog service: Manages book inventory (search by topic, fetch details).
- Order service: Processes purchases and updates stock.
- Frontend service: Acts as the API gateway for client requests.

We use Node.js with Express.js for our microservices because it's lightweight, scales easily, and efficiently handles asynchronous operations. Express simplifies routing and middleware while

keeping our services fast and maintainable. The large Node.js ecosystem and active community provide excellent support for development.

We use Docker to test the project, by follow these steps:

- Build three images.
 - Catalog-service.

```
PS C:\Users\USER-Q\Desktop\Bazar.com> docker build -t catalog-service .
```

- Order-service.

```
PS C:\Users\USER-Q\Desktop\Bazar.com> docker build -t order-service .
```

- Frontend-service.

```
PS C:\Users\USER-Q\Desktop\Bazar.com> docker build -t frontend-service .
```

- Build Bazar.com network.

```
PS C:\Users\USER-Q\Desktop\Bazar.com> docker network create proj-net
```

- Run all services.

```
PS C:\Users\USER-Q\Desktop\Bazar.com> docker-compose up --build
```

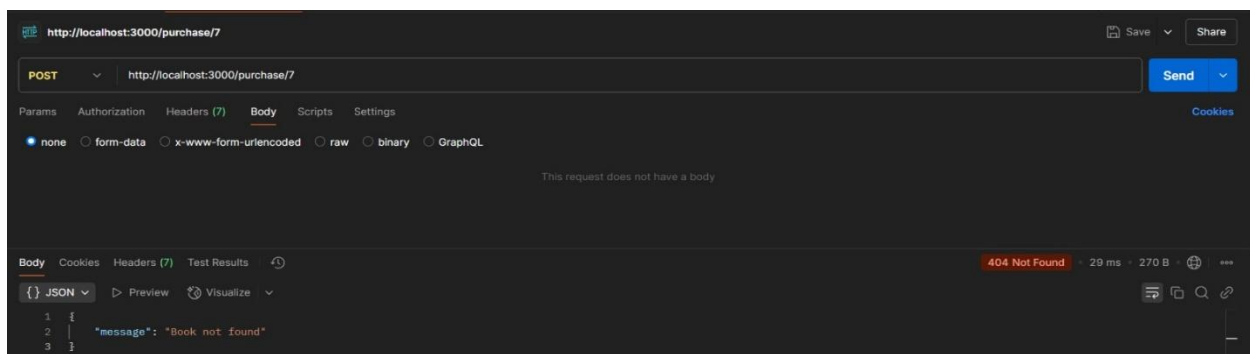
- Test endpoints in Postman.

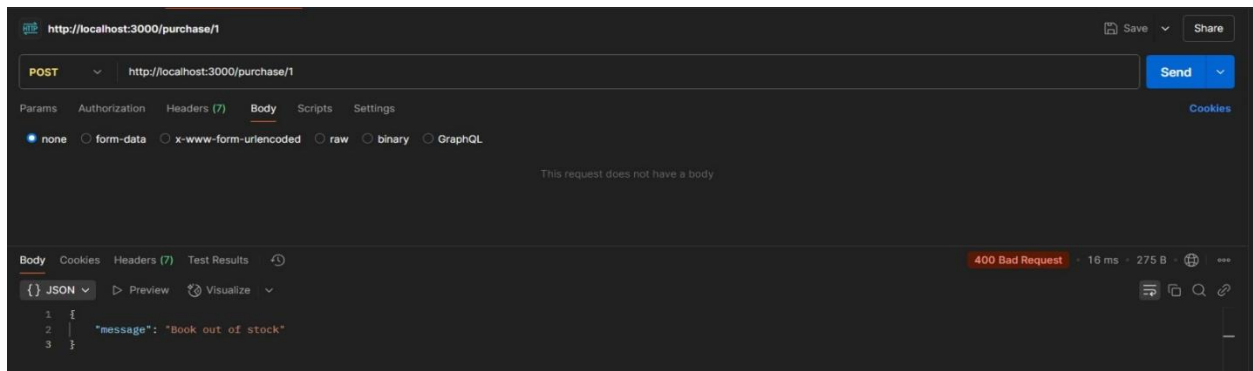
GET <http://localhost:5000/search/distributed%20systems> (Frontend → Catalog).

POST <http://localhost:5000/purchase/1> (Frontend → Order → Catalog).

Firstly, Catalog service receives requests from frontend service to search by topic or send information by id for a book. Also, it receives from order service to update quantity.

Secondly, Order service receives requests from frontend service to deal with purchase operation that goes to catalog service to modify quantity. It sends message “Book not found” if the book id not in the Bazar.com, or “Book out of stock” if there are no more copies of the book, or “book purchased successfully”





Finally, Frontend service it sends requests (search by topic and information for a book by id) for catalog service, or request (purchase) for order service.

